

BRYAN KUMARA

<https://brykumara.github.io/>

PERSONAL DATA

Place of Birth: Singapore

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BIOGRAPHY

I am a DPhil student at the University of Oxford where I am advised by Prof. Kasper Rasmussen researching applied cryptography for distributed systems. Before Oxford, I was a cryptographer at CrossBar Inc. working on threshold ECDSA and a researcher at the Alan Turing Institute. In the latter, I co-authored the Mahi-Mahi asynchronous DAG consensus protocol, a batched vector commitment for MPCitH-type signatures, and guidelines for implementing homomorphic encryption.

EXPERIENCES

- 2025 - **Applied Cryptographer**, CrossBar Inc., United States
Cryptography and Blockchain Team
- 2023 - 2025 **Research Assistant**, The Alan Turing Institute, United Kingdom
Privacy-Enhancing Technologies for Identity Systems
- 2021 - 2021 **Project Management Intern**, PayPal, United Kingdom
European Union Transition Project
- 2019 - 2019 **Undergraduate Researcher**, Imperial College London, United Kingdom
Model Checking Probabilistic Multi-Agent Systems
- 2018 - 2018 **Project Intern**, Boston Consulting Group, Indonesia
Indonesian Timber and Tissue Paper Manufacturing
- 2017 - 2017 **Research Assistant**, The Habibie Center, Indonesia
ASEAN Cybersecurity Policies

EDUCATION

- 2025 - **University of Oxford**, Oxford, United Kingdom
D.Phil in Computer Science
- 2021 -2022 **University College London**, London, United Kingdom
M.Sci in Emerging Digital Technologies
- 2017 - 2020 **London School of Economics**, London, United Kingdom
B.Sci in Philosophy, Logic, and Scientific Method

PUBLICATIONS AND REPORTS

My full list of publications and technical reports is available at <https://brykumara.github.io/>